

Course Code:B20BS2101																								
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)								R 20																
II B. Tech I Semester Regular Examinations																								
NUMERICAL METHODS & VECTOR CALCULUS																								
(Common to CE, CSE, EEE & IT)																								
Time: 3 Hrs.					Max. Marks:70																			
Answer any one Question from Each Unit																								
All questions carry equal Marks																								
	UNIT-I							CO	KL	M														
1.a)	Determine a real root of the equation $x \log_{10} x = 1.2$ by Regula-falsi method and correct to two decimal places							CO1	K2	7														
b)	Determine the cube root of 41 using Newton-Raphson method							CO1	K2	7														
(OR)																								
2. a)	Determine Newton's forward difference interpolation formula find Y (3), from the following table							CO1	K2	7														
	<table><tr><td>X</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>Y</td><td>7</td><td>11</td><td>14</td><td>18</td><td>24</td><td>32</td></tr></table>										X	0	5	10	15	20	25	Y	7	11	14	18	24	32
X	0	5	10	15	20	25																		
Y	7	11	14	18	24	32																		
b)	Using Lagrange's interpolation formula find Y (10) from the following table							CO1	K2	7														
	<table><tr><td>X</td><td>5</td><td>6</td><td>9</td><td>11</td></tr><tr><td>Y</td><td>12</td><td>13</td><td>14</td><td>16</td></tr></table>										X	5	6	9	11	Y	12	13	14	16				
X	5	6	9	11																				
Y	12	13	14	16																				
UNIT-II																								
3.a)	Evaluate $\int_0^1 x^3 dx$ with five subintervals by trapezoidal rule							CO2	K3	7														
b)	Evaluate $\int_0^2 \frac{dx}{x^3+x+1}$ by using Simpsons 1/3 rd rule with $h = 0.25$							CO2	K3	7														
(OR)																								
4. a)	Employ Taylor's method to obtain approximate value of y at $x=0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x, y(0) = 0$							CO2	K3	7														
b)	Evaluate $y(0.2)$ using Runge-Kutta 4 th order method, given $\frac{dy}{dx} = \frac{y^2-x^2}{y^2+x^2}, y(0) = 1.$							CO2	K3	7														
UNIT-III																								
5.a)	Apply change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dx dy$							CO3	K3	7														
b)	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates							CO3	K3	7														
(OR)																								

6. a)	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dx dy dz$	CO4	K3	7
b)	Determine the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$	CO4	K3	7
UNIT-IV				
7. a)	Obtain the directional derivative of $\phi = xy + yz + zx$ at A in the direction of AB where $A = (1,2,-1)$, $B = (5,6,8)$.	CO5	K2	7
b)	Determine the values of a and b such that the surface $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1,-1,2)$.	CO5	K2	7
(OR)				
8. a)	Show that the vector $(x^2 - yz)\bar{i} + (y^2 - zx)\bar{j} + (z^2 - xy)\bar{k}$ is irrotational and find its scalar potential.	CO5	K2	7
b)	Determine $\text{Curl } \bar{F}$ and $\text{div } \bar{F}$ for $\bar{F} = x^2y\bar{i} - 2xz\bar{j} + 2yz\bar{k}$	CO5	K3	7
UNIT-V				
9.a)	Determine the work done in moving a particle once round the circle $x^2 + y^2 = 9$ in the xy- plane by the force $\bar{F} = (2x - y - z)\bar{i} + (x + y - z^2)\bar{j} + (3x - 2y + 4z)\bar{k}$	CO6	K3	7
b)	Evaluate the line integral by Stokes's theorem for the vector function $\bar{F} = y^2\bar{i} + x^2\bar{j} + (z+x)\bar{k}$ and C is the triangle with vertices $(0,0,0)$, $(1,0,0)$ and $(1,1,0)$.	CO6	K3	7
(OR)				
10.	Verify Green's theorem in the plane For $\oint_C [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$, where C is boundary of the region defined by $y = \sqrt{x}$, $y = x^2$	CO6	K3	14

				Course Code: B20 CS 2101		
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20		
II B.Tech I Semester Regular Examinations						
OBJECT ORIENTED PROGRAMMING THROUGH C++						
(Computer Science & Engineering)						
Time: 3 Hrs.				Max. Marks:70		
Answer any one Question from Each Unit						
All questions carry equal Marks						
				CO	KL	M
UNIT-I						
1	a.	Discuss the differences between C and C++.	CO1	K2	7	
	b.	Explain about disadvantages of conventional programming.	CO1	K2	7	
OR						
2	a.	Explain about key concepts of Object Oriented Programming.	CO1	K2	8	
	b.	Explain about advantages of OOPS.	CO1	K2	6	
UNIT-II						
3	a.	Explain Classes, Objects and Member Functions.	CO2	K2	6	
	b.	Write a C++ program to overload area() and perimeter() function to calculate area of shapes like triangle, square, circle and rectangle.	CO2	K3	8	
OR						
4	a.	How will you destroy the objects initialized by the constructor in the program?	CO2	K2	5	
	b.	Explain the use of different constructors (default, parameterized and copy constructors) with suitable examples.	CO2	K2	9	
UNIT-III						
5	a.	Explain inheritance with the advantages and disadvantages.	CO3	K2	8	
	b.	Illustrate the visibility of base class members for the access specifiers: private, protected and public while creating the derived class and also explain the syntax for creating derived class.	CO3	K3	6	
OR						
6	a.	What are the various types of situations that might arise and can be handled in data conversion between incompatible types?	CO3	K2	7	
	b.	Write C++ Program to overload + operator to add two matrices.	CO3	K3	7	
UNIT-IV						
7	a.	Explain virtual classes and their need while building class hierarchy.	CO4	K2	7	
	b.	Explain the role of this pointer in C++ with a programming example.	CO4	K2	7	
OR						
8	a.	How does polymorphism promote extensibility? Illustrate.	CO4	K3	6	
	b.	With a program explain how late binding can be achieved in C++.	CO4	K3	8	
UNIT-V						
9	a.	Explain Class Template and Function Template	CO5	K2	8	
	b.	Write a C++ program that illustrates exception handling with the help of keywords: try, throws and catch.	CO5	K3	6	
OR						
10	a.	What is STL? Briefly explain the use of containers, vectors, lists and maps.	CO5	K2	7	
	b.	Write a C++ program for Generic Bubble Sort using Template Functions.	CO5	K3	7	

B20CS2102						
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20		
II B. Tech I Semester Regular Examinations						
SOFTWARE ENGINEERING						
(Computer Science & Engineering)						
Time: 3 Hrs.				Max. Marks:70		
Answer any one Question from Each Unit						
All questions carry equal Marks						
				CO	KL	M
UNIT-I						
1	a.	Explain about Nature of the software.	CO1	K2	7	
	b.	Explain about Evolutionary process models.	CO1	K2	7	
OR						
2	a.	Explain about Software Myths.	CO1	K2	7	
	b.	Explain about Agile Process.	CO1	K2	7	
UNIT-II						
3	a.	Explain the concept of use cases with the help of an example.	CO2	K2	7	
	b.	Explain the process of requirements engineering.	CO2	K2	7	
OR						
4	a.	State and explain various aspects in the requirements validation process.	CO2	K2	7	
	b.	Elaborate the main focus of requirement analysis.	CO2	K2	7	
UNIT-III						
5	a.	Explain about UML models that supplement usecases.	CO3	K2	7	
	b.	What is Class based Modeling? Explain Elements of Class based Modeling.	CO3	K2	7	
OR						
6	a.	Explain the steps to create a behavior model for a System.	CO3	K2	7	
	b.	Explain about the requirements patterns with an example.	CO3	K2	7	
UNIT-IV						
7	a.	Explain about different architecture Styles.	CO4	K2	8	
	b.	Explain Web Apps Interface Design.	CO4	K2	6	
OR						
8	a.	Explain the process of Designing Class-Based Components for WebApps	CO4	K2	8	
	b.	Explain the Golden Rules for User Interface Design	CO4	K2	6	
UNIT-V						
9	a.	Explain the testing strategies for conventional software.	CO5	K2	7	
	b.	What is Debugging? Explain about the art of Debugging.	CO5	K2	7	
OR						
10	a.	Explain the testing strategies for Object Oriented software.	.CO5	K2	7	
	b.	Explain basic path testing with one example	CO5	K2	7	

				Course Code: B20 CS 2103		
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20		
I B. Tech I Semester Regular Examinations						
OPERATING SYSTEMS						
(Computer Science & Engineering)						
Time: 3 Hrs.				Max. Marks:70		
Answer any one Question from Each Unit						
All questions carry equal Marks						
				CO	KL	M
UNIT-I						
1	a.	Explain the abstract view of system components.	CO1	K2	7	
	b.	Discuss the Simple Operating System Structure.	CO1	K2	7	
OR						
2	a.	Explain different types of Operating Systems.	CO1	K2	7	
	b.	Define a System call. Explain the various types of system calls provided by Operating System,	CO1	K2	7	
UNIT-II						
3	a.	Differentiate one- to- one, many- to-one multi threading models.	CO2	K2	7	
	b.	Explain Dining Philosophers problem? Discuss the solution to Dining Philosopher's problem using monitors.	CO2	K2	7	
OR						
4	a.	Explain Primitive Priority Scheduling Algorithms with an Example?	CO2	K2	7	
	b.	Discuss the solution to Reader/Writers Problem using semaphores.	CO2	K2	7	
UNIT-III						
5	a.	Differentiate paging and segmentation.	CO3	K2	7	
	b.	Explain briefly the performance of Demand paging with an example.	CO3	K2	7	
OR						
6	a.	Define Page Fault. When does a page fault occur? Describe the action taken by OS when page fault occurs.	CO3	K2	7	
	b.	Apply FIFO and LRU page replacement algorithms for the following string to determine the number of page faults. 7 0 1 2 0 3 0 4 2 3 0 2 1 2 0 1 7 0 1 for a memory with '3' frames.	CO3	K3	7	
UNIT-IV						
7	a.	Apply the deadlock detection algorithm to determine deadlock will exist or not for the following system with 5 process and 3 resource types (resource type A has 7 instances, B has 2 instances, and C has 6 instances) Snapshot at time T0 Process Allocation Request Available A B C A B C A B C P0 0 1 0 0 0 0 0 0 0 P1 2 0 0 2 0 2 P2 3 0 3 0 0 0 P3 2 1 1 1 0 0 P4 0 0 2 0 0 2	CO4	K3	7	
	b.	Explain various File access methods with Suitable examples	CO4	K2	7	
OR						

8	a.	Explain deadlock avoidance using banker's algorithm with suitable example.	C04	K2	7
	b.	Apply FCFS, SSTF disk arm scheduling schemes to find total number head movements for the following string 98 183 37 122 14 124 65 67 assume the head pointer at 53.	C04	K3	7
UNIT-V					
9	a.	Explain System and Network Threats	C05	K2	7
	b.	Describe the System Component of Windows XP architecture	C05	K2	7
OR					
10	a.	Explain Principles and domain Protections.	C05	K2	7
	b.	Describe the components of the Linux System.	C05	K2	7

Course Code: B20 CS 2104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					
R 20					
I B. Tech I Semester Regular Examinations					
PYTHON PROGRAMMING					
(Computer Science & Engineering)					
Time: 3 Hrs.		Max. Marks:70			
Answer any one Question from Each Unit					
All questions carry equal Marks					
		CO	KL	M	
UNIT-I					
1	a.	Explain the basic data types available in Python with examples.	C02	K1	6
	b.	List and explain all the operators available in python.	C02	K1	8
OR					
2	a.	Illustrate the different types of control flow statements in Python	C02	K1	8
	b.	Write a program to display the fibonacci sequence up to nth term where n is provided by the user.	C03	K1	6
UNIT-II					
3	a.	Write Python program to count the total number of vowels, consonants and blanks in a String.	C03	K2	6
	b.	Compare mutable and Immutable data types in Python with suitable examples.	C02	K2	8
OR					
4	a.	Explain List operations available in python.	C02	K2	8
	b.	Write a program to Display both the keys and values sorted in alphabetical order by the key.	C03	K2	6
UNIT-III					
5	a.	Explain the concept of accessibility of variables in nested functions.	C02	K3	8
	b.	Define a function that calculates the sum of all numbers from 0 to its argument.	C03	K3	6
OR					
6	a.	What are modules, how do you use them in your programs.	C02	K3	6
	b.	Write a short note on i) Keyword Arguments ii) Lambda Functions.	C02	K3	8
UNIT-IV					
7	a.	Discuss the following methods associated with the file object a) read() b) readline() c) readlines() d) tell() e) seek() f) write().	C02	K4	8
	b.	Write Python Program to Count the Number of Words in a given file.	C03	K4	6
OR					
8	a.	Explain the concept of class methods and static methods with examples.	C02	K4	6
	b.	Write Python Program to Demonstrate Multiple Inheritance with Method Overriding.	C03	K4	8
UNIT-V					
9	a.	What is the difference between else block and finally block in exception handling? Explain with an example program.	C03	K5	7
	b.	Discuss in detail about Scrolling list boxes.	C02	K5	7
OR					
10	a.	Explain in detail about User defined exceptions.	C02	K5	7
	b.	Explain about command buttons and Responding to events.	C02	K5	7

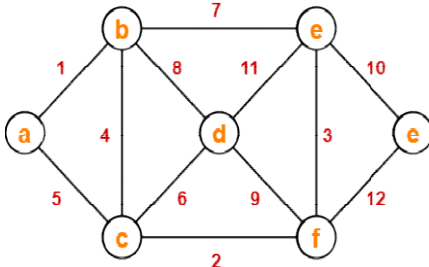
Course Code:B20BS2101																									
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20																					
II B. Tech I Semester Regular Examinations																									
PROBABILITY AND STATISTICS																									
(Computer Science & Engineering)																									
Time: 3 Hrs.		Max. Marks:70																							
Answer any one Question from Each Unit																									
All questions carry equal Marks																									
Q.No	Questions	CO	KL	M																					
1.a)	What are the different methods in collecting Primary and Secondary data?	CO1	K2	7																					
b)	Lives of two models of Refrigerators in a research survey are given below. <table border="1"><tr><td>Lives in Year</td><td>0 -2</td><td>2 - 4</td><td>4 - 6</td><td>6 - 8</td><td>8 -12</td><td>12 -14</td></tr><tr><td>Model - A</td><td>2</td><td>16</td><td>13</td><td>7</td><td>5</td><td>4</td></tr><tr><td>Model - B</td><td>2</td><td>7</td><td>12</td><td>19</td><td>9</td><td>1</td></tr></table> Determine the average life of each model and which model has greater uniformity.	Lives in Year	0 -2	2 - 4	4 - 6	6 - 8	8 -12	12 -14	Model - A	2	16	13	7	5	4	Model - B	2	7	12	19	9	1	CO1	K3	7
Lives in Year	0 -2	2 - 4	4 - 6	6 - 8	8 -12	12 -14																			
Model - A	2	16	13	7	5	4																			
Model - B	2	7	12	19	9	1																			
(OR)																									
2.a)	Define Random variable and explain about types of random variables with examples. Also discuss discrete and continuous probability functions of random variables.	CO1	K2	7																					
b)	A continuous random variable X has a pdf $f(x) = 3x^2, 0 < x < 1$. Determine a and b such that (i) $P(X \leq a) = P(X > a)$ and (ii) $P(X > b) = 0.05$.	CO1	K3	7																					
3.a)	A two-dimensional r.v. (X, Y) has a bivariate distribution given by $P(X, Y) = \frac{x^2 + Y}{32}, for x = 0, 1, 2, 3 and y = 0, 1$. Calculate marginal distributions of X and Y.	CO2	K3	7																					
b)	Let X be a random variable with the following distribution <table border="1"><tr><td>X</td><td>6</td><td>9</td><td>12</td></tr><tr><td>P(X=x)</td><td>1/6</td><td>1/2</td><td>1/3</td></tr></table> Determine E(x), E(x ²) and using the laws of expectation calculate E(2X+1) ²	X	6	9	12	P(X=x)	1/6	1/2	1/3	CO2	K3	7													
X	6	9	12																						
P(X=x)	1/6	1/2	1/3																						
(OR)																									
4. a)	A random variable X has the probability density function given by $f(x) = \begin{cases} x & \text{if } 0 \leq x < 1 \\ 2 - x & \text{if } 1 \leq x < 2 \\ 0 & \text{elsewhere} \end{cases}$ Determine E(X) and V(X).	C02	K3	7																					
b)	Define moment generating function and write its properties	C02	K2	7																					

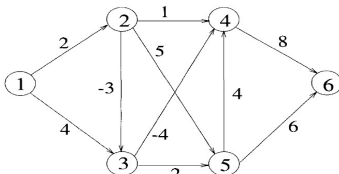
5.a)	Determine a 2 nd degree regressed polynomial $y = a + b x + c x^2$ for the following data						CO3	K3	7				
	x	0	1	2	3	4							
	y	1	1.5	2.6	4.2	6.8							
b)	Determine the regression lines of x on y for the following data						CO3	K3	7				
	X	1	2	3	4	5				6			
	Y	15	17	14	18	16				15			
(OR)													
6.a)	Describe the procedure for fitting an exponential curve of the form $y = ax^b$						CO4	K3	7				
b)	Calculate the rank correlation between male and female ranks in a game						CO4	K2	7				
	Male ranks		3	5	1	4				2	7	6	8
	Female ranks		4	2	3	1				5	6	8	7
7.a)	Ten coins are thrown simultaneously. Determine the probability of getting at least i) seven heads ii) six heads						C05	K3	7				
b)	Fit a Poisson distribution for the following data						C05	K2	7				
	x	0	1	2	3	4				5	6	7	8
	f(x)	56	156	132	92	37				22	4	0	1
(OR)													
8. a)	In a normal distribution 31% of the items are under 45 and 8% are over 64. Determine mean and standard deviation						C05	K3	7				
b)	Derive mean and variance of Uniform distribution						C05	K3	7				
9.a)	Explain the following concepts						C06	K2	7				
	(i) Null and Alternative Hypothesis (ii) Type I and Type II errors (iii) Critical region and level of significance.												
b)	Before an increase in excise duty on tea, 800 persons out of a sample of 1,000 persons were found to be tea drinkers. After an increase in duty, 800 people were tea drinkers in a sample of 1,200 people. Using standard error proportion, state whether there is a significant decrease in the consumption of tea after the increase in excise duty.						C06	K2	7				
(OR)													
10.a)	The height of 10 males of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches, Is it reasonable to believe that the average height is greater than 64 inches? Test at 5% significance level assuming that for 9 degrees of freedom $P(t > 1.83) = 0.05$.						C06	K2	7				

b)	The following figures show the distribution of digits in numbers chosen at random from a telephone directory.										C06	K3	7	
	Digits	0	1	2	3	4	5	6	7	8				9
	frequency	1026	1107	997	966	1075	933	1107	972	964				853
	Test whether the digits may be taken to occur equally frequently in the directory.													

Course Code:B20 CS 2201							
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20		
I B. Tech II Semester Regular Examinations							
DATABASE MANAGEMENT SYSTEMS							
(Computer Science & Engineering)							
Time: 3 Hrs.			Max. Marks:70				
Answer any one Question from Each Unit							
All questions carry equal Marks							
					CO	KL	M
UNIT-I							
1	a.	Compare Database Management Systems with File Processing Systems	CO1	K3	8		
	b.	Explain the roles of different database users	CO1	K2	6		
OR							
2	a.	Discuss the applications of Database Management Systems	CO1	K2	6		
	b.	Describe the structure of a Database Management System	CO1	K2	8		
UNIT-II							
3		Give syntax and apply the SQL commands for defining two example tables of your choice. Then insert data, update data in the tables	CO2	K3	14		
OR							
4		What are relational instances and schemas? How'd you use keys and schemas in relational model?	CO2	K3	14		
UNIT-III							
5	a.	Apply conceptual DB design and draw E-R diagram for the following situations by assuming appropriate Attributes i) A Part is supplied by many suppliers at different costs and a supplier supplies many parts ii) An employee works in at most one department and a department has many employees iii) A house has at least one and at most one owner and owner has many houses iv) A muslim woman marries at most one man and a muslim man could marry many women	CO3	K3	8		
	b.	Demonstrate set operations in SQL	CO2	K3	6		
OR							
6	a.	Apply different kinds of joins in SQL to example queries	CO2	K3	7		
	b.	Illustrate basic features of ER model	CO3	K3	7		
UNIT-IV							
7	a.	Apply Loss-less join decomposition into BCNF for an example table	CO4	K3	8		
	b.	Apply dependency preserving decomposition into 3NF for an example table	CO4	K3	6		
OR							
8		Illustrate Normal forms from 1 NF to BCNF.	CO4	K3	14		

UNIT-V					
9		What are page tables and Transaction tables? Describe analysis, redo and undo steps of ARIES.	CO5	K2	14
OR					
10		Explain 2PL and time stamp ordering protocols	CO5	K2	14

Course Code: B20 CS 2202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R 20		
II B.Tech II Semester Regular Examinations					
DESIGN ANALYSIS OF ALGORITHMS					
(Computer Science & Engineering)					
Time: 3 Hrs.		Max. Marks:70			
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT-I					
1	a.	Define an algorithm. What are the different criteria that satisfy the algorithm? (algorithm specifications)	CO1	K2	6
	b.	What is Strassen's matrix multiplication? Explain its time complexity?	CO1	K3	8
OR					
2	a.	Explain how algorithms performance is analysed? Describe asymptotic notations?	CO1	K2	6
	b.	Draw the tree of calls of merge sort for the following set. (35, 25,15,10,45, 75, 85, 65, 55, 5, 20, 18)	CO1	K3	8
UNIT-II					
3	a.	Present a Greedy Algorithm for Sequencing Unit Time Jobs with deadlines and profits.	CO2	K3	6
	b.	Illustrate , how to find the minimum cost spanning tree by using Prim's Algorithm	CO2	K3	8
OR					
4	a.	Solve knapsack problem with greedy approach for instances n=3, m=6, profits are (p1, p2, p3) =(1,2,5), weights are(w1,w2,w3)=(2,3,4).	CO2	K3	6
	b.	What is a Minimum cost spanning tree? Identify an efficient data structure for implementation of Kaushal's Algorithm for the connected Weighted Graph. 	CO2	K3	8
UNIT-III					
5	a.	Construct an optimal travelling sales person tour using Dynamic Programming. 0 10 9 3 5 0 6 2 9 6 0 7 7 3 5 0	CO3	K2	7
	b.	Apply dynamic programming to obtain optimal binary search tree for the identifier set (a1, a2, a3, a4)=(cin, for, int, while) with (p1, p2, p3, p4)=(1, 4, 2, 1), (q0, q1, q2, q3, q4)=(4, 2, 4, 1, 1) and also write algorithm for its construction.	CO3	K3	7
OR					

6	a.	Solve 0/1 knapsack problem for the following data using sets method Let us consider that the capacity of the knapsack is $W = 25$ and the items are as shown in the following table. <table><thead><tr><th>Item</th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><td>Profit</td><td>24</td><td>18</td><td>18</td><td>10</td></tr><tr><td>Weight</td><td>24</td><td>10</td><td>10</td><td>7</td></tr></tbody></table>	Item	A	B	C	D	Profit	24	18	18	10	Weight	24	10	10	7	C03	K3	7
Item	A	B	C	D																
Profit	24	18	18	10																
Weight	24	10	10	7																
	b.	Find the shortest path from node 1 to every other node in the graph given below using all pairs shortest algorithm. 	C03	K3	7															
UNIT-IV																				
7	a.	Relate Hamiltonian cycle with travelling sales person problem and also give the backtracking solution vector that finds all Hamiltonian cycles for any directed or undirected graph.	C04	K3	8															
	b.	Write the branch and bound algorithm to generate minimum length tour for the given cost adjacency matrix. $\begin{bmatrix} \infty & 18 & 28 & 8 & 9 \\ 13 & \infty & 14 & 2 & 1 \\ 1 & 3 & \infty & 1 & 2 \\ 17 & 4 & 16 & \infty & 1 \\ 14 & 2 & 5 & 16 & \infty \end{bmatrix}$	C04	K2	6															
OR																				
8	a.	Draw the portion of state space tree generated by recursive backtracking algorithm for sum of subsets problem with an example.	C04	K3	8															
	b.	Consider the knapsack instance $n=4$, $(p_1, p_2, p_3, p_4)=(10, 10, 12, 18)$, $(w_1, w_2, w_3, w_4)=(2, 4, 6, 9)$ and $m=15$. Solve this 0/1 knapsack problem using least cost branch and bound	C04	K3	6															
UNIT-V																				
9	a.	Write short notes on i) Classes of NP-hard ii) Classes of NP-complete and Prove that if $NP \neq CO - NP$, then $P \neq NP$	C05	K2	8															
	b.	Draw the Decision Tree and Find the number of Key Comparison in the worst and average case for: a. The four Element Binary search b. the Three-element basic insertion sort	C05	K2	6															
OR																				
10	a.	Explain P, NP-hard, NP-complete problems and their relationship using pi diagram. Also write the nondeterministic polynomial time algorithms for clique, 0/1 knapsack and satisfiability problems.	C05	K3	8															
	b.	Explain lower bound Arguments in detail	C05	K2	6															

				Course Code: B20 CS 2203		
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20		
II B.Tech II Semester Regular Examinations						
JAVA PROGRAMMING						
(Computer Science & Engineering)						
Time: 3 Hrs.				Max. Marks:70		
Answer any one Question from Each Unit						
All questions carry equal Marks						
				CO	KL	M
UNIT-I						
1	a.	Explain about the Features of Java.		CO1	K2	7
	b.	Explain about Method Overloading with an example.		CO1	K2	7
OR						
2	a.	Explain various types of Constructors with suitable examples.		CO1	K2	8
	b.	Explain with an example about this keyword.		CO1	K2	6
UNIT-II						
3	a.	Explain the differences between Arrays and Vectors.		CO2	K2	7
	b.	Write a Java Program to implement methods of Vector class.		CO2	K3	7
OR						
4	a.	Explain the differences between String class and StringBuffer class.		CO2	K2	6
	b.	Write a Java Program to implement various methods of String class.		CO2	K2	8
UNIT-III						
5	a.	Explain different types of inheritances in Java.		CO3	K2	8
	b.	Explain with example about Method Overriding.		CO3	K3	6
OR						
6	a.	Differentiate Abstract class and Interface.		CO3	K4	6
	b.	Discuss about procedure for creating packages with an example.		CO3	K2	8
UNIT-IV						
7	a.	Explain about the mechanism of Exception handling in Java.		CO4	K2	7
	b.	Write a Java Program to create Custom Exception.		CO4	K2	7
OR						
8	a.	Explain about the life cycle of thread.		CO4	K2	7
	b.	Explain about the various methods of creating Thread.		CO4	K2	7
UNIT-V						
9	a.	Discuss about Byte Oriented IO and Character Oriented IO.		CO5	K2	7
	b.	Write a Java program to copy contents of one file to another file using Line Oriented IO.		CO5	K2	7
OR						
10	a.	Discuss the types of JDBC Drivers.		CO5	K2	8
	b.	Write a JDBC program to retrieve data from the database.		CO5	K2	6

Course Code: B20HS2201				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R 20	
II B. Tech II Semester –MODEL PAPER				
MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY				
Computer Science & Engineering				
Time : 3Hrs.			Max. Marks: 70	
Answer any one Question from EACH UNIT.				
All questions carry equal marks.				
S.No.	Question / Problem	CO	KL	M
	UNIT-I			
1.	Define Managerial Economics and Explain its nature and scope	CO1	K2	14
	OR			
2.	What do you mean by Elasticity of demand? Explain in detail about degrees of Price elasticity of Demand?	CO1	K2	14
	UNIT-II			
3.	Define Cost & classify the Elements of Cost?	CO2	K2	14
	OR			
4.	How do you calculate BEP? What are its Assumptions and Applications?	CO2	K3	14
	UNIT-III			
5.	What are Market Structures and explain the features of Perfect Competition?	CO3	K2	14
	OR			
6.	Why is pricing significant in the context of business? Describe any four pricing practices?	CO3	K2	14
	UNIT-IV			
7.	Describe about the Importance of Accounting and types of accounts	CO4	K2	14
	OR			
8.	From the following Trail Balance of Suresh as at December 31, 2013, prepare Trading, Profit and Loss Account for the year ended December 31, 2013 and a Balance Sheet as on that date:	CO4	K3	14

	Dr. (Rs.)	Cr. (Rs.)			
Purchases of materials	32,000				
Productive wages	13,000				
Sales		60,000			
Salaries	4,000				
Travelling expenses	1,000				
Carriage inwards	550				
Insurance	300				
Commission	650				
Rent and rates	1,000				
Cash in hand	350				
Cash at bank	5,550				
Repairs	600				
Sundry expenses	110				
Mortgage		6,100			
Buildings	8,000				
Machinery	3,000				
Furniture	1,000				
Stock on hand (1.1.2013)	11,500				
Capital		21,310			
Sundry debtors	9,000				
Sundry creditors		4,200			
	91,610	91,610			

	Adjust the following: (a) Prepaid rent Rs. 100 (b) Depreciate the following: • Buildings @ 10per cent per annum • Machinery @ 20 per cent per annum • Furniture @ 15 per cent per annum (c) Provide for bad debts Rs. 100 (d) Outstanding insurance Rs. 50 (e) Closing stock Rs. 12,000			
	UNIT-V			
9.	Explain about capital and the sources available for raising finance	CO5	K2	14
	OR			
10.	Explain about the concept and causes of depreciation. Evaluate the straight-line method and diminishing balance methods.	CO5	K2	14